Projekt 4N: Nordwest Niedersachsen Nachhaltig Neu
TV-7 Landwirtschaftlicher Strukturwandel und Klimawandel-Anpassung

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Wind power estimation

Internal Report, Technical Report

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Contents

1 Introduction

Wind power is probably the most important source for renewable energy in Germany. The aim of this work is to assess the power output of a wind turbine at a given location. The starting point for the calculation is the wind speed at a height of 10 m, which can be obtained from meteorological data. The source for this work is the PVGIS website [1], which provides radiation data for solar power and wind speed.
Within this work a prediction model for wind turbines is tested for the wind turbine located at the Campus Emden. The model consists of two parts. At first, the wind speed is calculated for the value at hub height. Secondly, the power output is simulated by using a characteristic power curve. Finally, simulation data is compared with the measured power output for the years 2020 to 2023.

The documentation should contain the following files.

- *fit-profile.py*: Calculates wind speed at different heights (Logarithmic Wind Profile and Power Law).
- *fit-power.py*: Calculates the power curve for the wind turbine (fit to reference data).
- *wind-year.py*: Simulates wind power generation for one year based on meteorological data and compares this to measured data.
- *func.py*: Function declarations required by *wind-year.py*.
- *Timeseries_Emden_40kWp_2013_2023.csv*: Meteorological data for Campus Emden [1].
- *W-2020.dat*: Measured power output for the wind turbine in 2020. The data was corrected for server downtime and maintanance. Similiar files exist for 2021, 2022 and 2023.

2 Modeling procedure

The wind turbine at Campus Emden was installed in the year 1994. Table 1 lists the specific data.

Table 1: Reference data of the wind turbine [2, 3].

Model type	Enercon E-18
Rated power	80 kW
Tower height	34 m
Hub height	36 m
Rotor diameter	18 m

Calculating the wind speed at hub height

The wind speed from meteorological data is given for a height of 10 m. This needs to be converted to the wind speed at hub height, which is 36 m in this case. In general, there are two equations available. The *logarithmic height formula* is derived from the Boundary-Layer Theory [4–6]. It describes the fact that the speed of a fluid approaches almost zero close to a boundary, e.g. the wall of a pipe or in our case the ground, and reaches higher values further away from the boundary, e.g. in the middle of a pipe or in our case the wind above a certain height. Equation 1 describes the wind speed u at the target height h_2 as a function of the velocity at reference height h_1 and the surface roughness length r. In this case a roughness length of $r = 0.016$ m is used, which was fitted to measured data.

$$u(h_2) = u(h_1) * \frac{\ln(h_2/r)}{\ln(h_1/r)} \tag{1}$$

The parameter r is associated with surface roughness. It is used to scale the height in equation 1 in order to account for disturbance of the profile. It is a simple and effective way to correct the equation for the influence of obstacles such as buildings and tress. In reality, builings and trees also introduce turbulences to the air flow. Therefore, equation 1 is less accurate within this zone, which spans at least three times the height of the tallest object (building). Another approach to describe the wind profile near the surface is the *power law approximation* (also called Hellman Equation) [5–9]. Equation 2 gives the velocity at target height h_2 as a function of the wind speed at height h_1 and the exponent α. The so called wind shear exponent α accounts for the effect of surface roughness.

$$u(h_2) = u(h_1) * \left(\frac{h_2}{h_1}\right)^{\alpha} \tag{2}$$

Both parameters r and α can be obtained from tables, which provide a rough estimate of the values. They can also be calculated from each other by using equation 3. The value of $r = 0.016$ m corresponds to $\alpha = 0.130$. Table 2 gives a short overview about the range of values [5, 6], where α is calculated for a target height of 36 m.

$$\alpha = \frac{1}{\ln(h_2/r)} \tag{3}$$

Table 2: Roughness length r and calculated wind shear exponent α [5, 6].

r in m	α	Comment
1.0	0.279	City, forest
0.5	0.234	Suburbs
0.2	0.193	Built-up terrain
0.05	0.152	Agricultural terrain
0.01	0.122	Meadow, airports
0.001	0.095	Snow on smooth ground
0.0001	0.078	Large water surfaces

Figure 1 shows a comparison of the two equations for an initial velocity of 5 m/s at the height of 10 m. It can be recognized that the logarithmic law yields a little higher values than the power law.

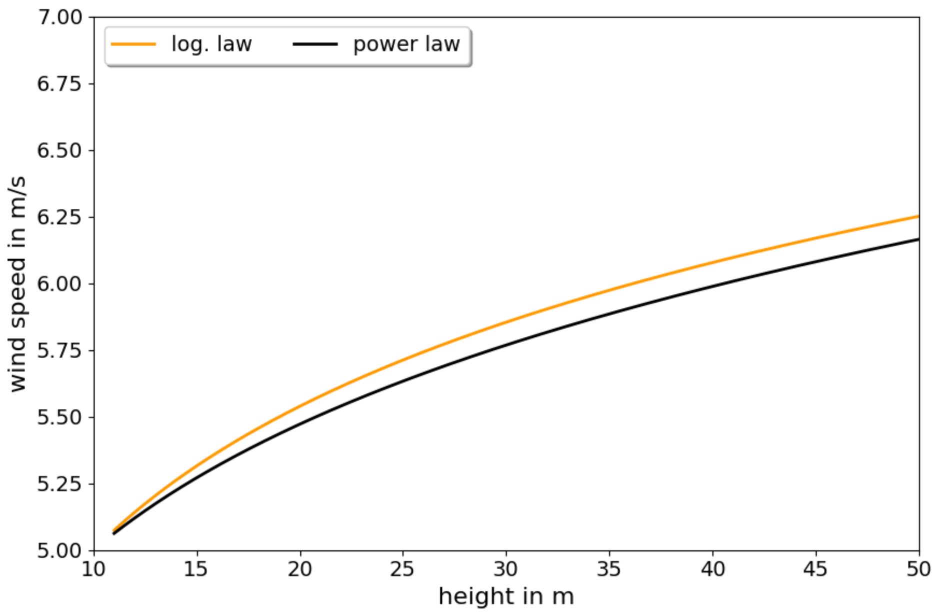

Figure 1: Calculated velocity profile 1 (logarithmic law) and equation 2 (power law).

Calculating turbine power

The power of the wind turbine is obtained from a power curve. In this case there are two sets of references. The first set comes from the manufacturer [2] and accounts for the turbine as installed in the year 1994. The second set comes from the internal documents at HS Emden-Leer [10]. Both sets differ slightly, because the turbine was modified during it's 30 years of lifetime. The fitted function for the power curve is given in equation 4. For values $p > 80\,\text{kW}$ the function is cut off.

$$p = 0.09 * u^{2.82} \tag{4}$$

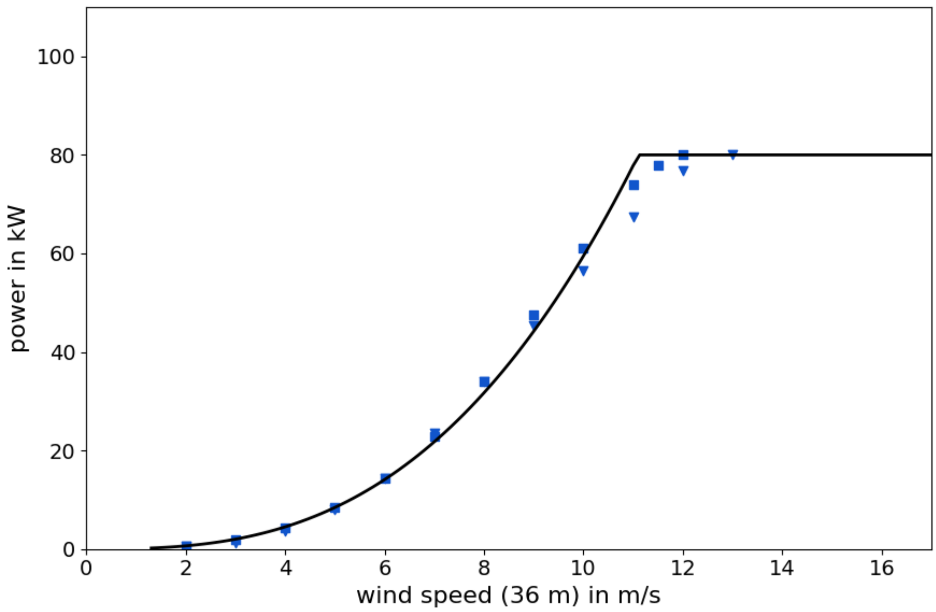

Figure 2: Power output for the wind turbine. For references: squares = HS Emden [10] and triangles = Enercon [2].

3 Results

For the following simulations the meterological data from the PVGIS website [1] was used for the years 2020 to 2023. The measured data is available at HS Emden-Leer. Unfortunatley, there were several shutdowns or server losses during that time. For the comparison of the data these times were excluded. Table 3 summarizes the results.

Table 3: Summary of measured and simulated data.

Year	Measured / kWh	Log. law / kWh	Power law / kWh
2020	97146.16	95535.27 (-2%)	91874.16 (-5%)
2021	93929.15	93996.34 ($\pm0\%$)	90420.60 (-4%)
2022	77844.55	82351.87 ($+6\%$)	79130.75 ($+2\%$)
2023	93105.80	103659.38 ($+11\%$)	99779.51 ($+7\%$)

The overall fit of the simulation data is satisfactory for a general power calculation. The deviation from measured data is about $\pm10\%$, which is not bad, given the fact that the equations for the vertical wind profile are aproximations. In general, the simulations based on the logarithmic profile yield somewhat higher values than those based on the power law, as can already be derived from figure 1. It should also be noted that the data for wind speed at the height of 10 m does not refer to the specific location at the Campus Emden, but is a general value for the city of Emden. The real wind speed at hub height of the wind power generator also depends on local turbulences caused by the surrounding buildings and therefore also on wind direction.

Overall, the model works well for anual simulations. For shorter periods a more detailed model is required. Figure 3 shows the data for July 3-4, 2020 as an example that the general trend is captured by the simulation, but the actual data may differ significantly. The year 2021 shows the best overall fit. Figure 4 shows the data for January 11 and figure 5 shows the data for July 27, 2021. The fit for these days is very good.

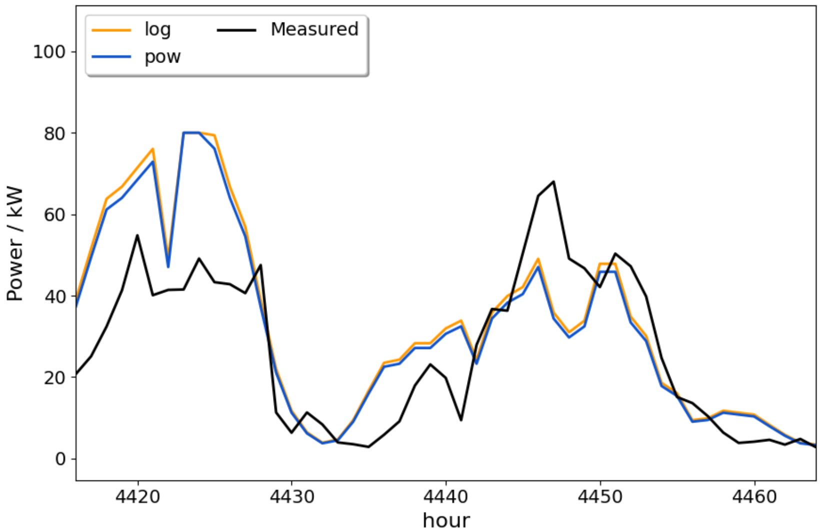

Figure 3: Wind power output for July 3-4, 2020.

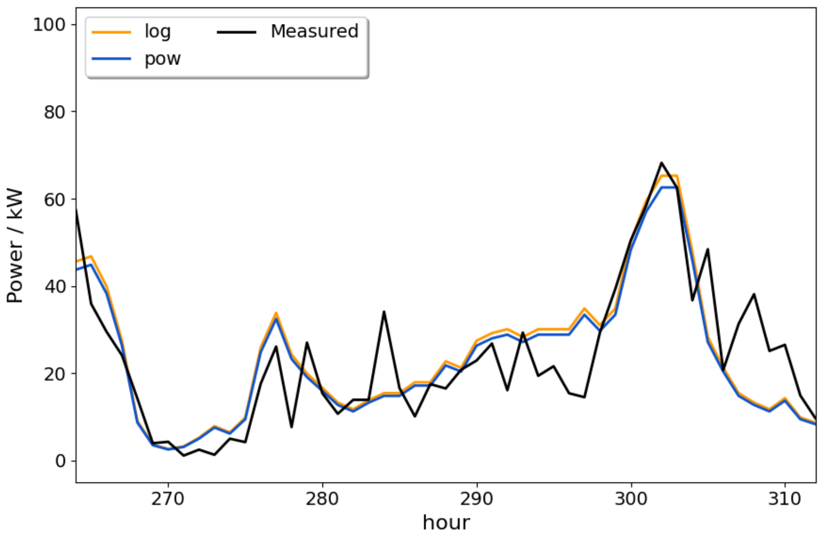

Figure 4: Wind power output for January 11-12, 2021.

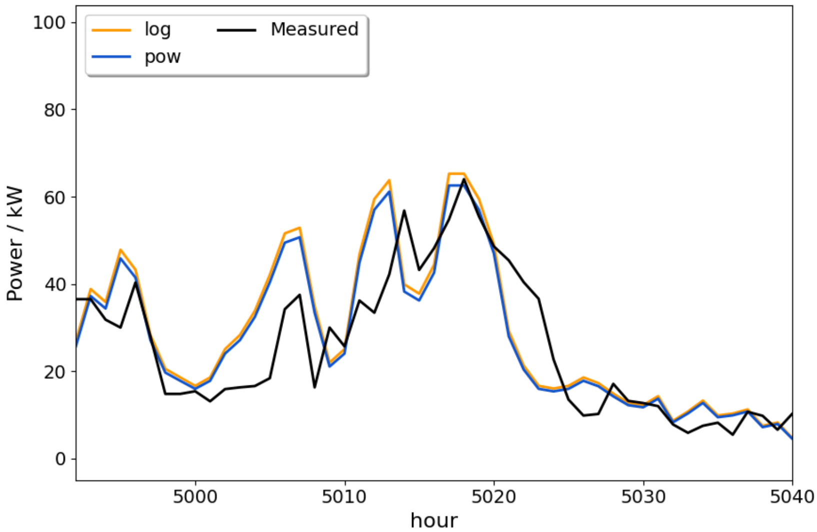

Figure 5: Wind power output for July 27-28, 2021.

References

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